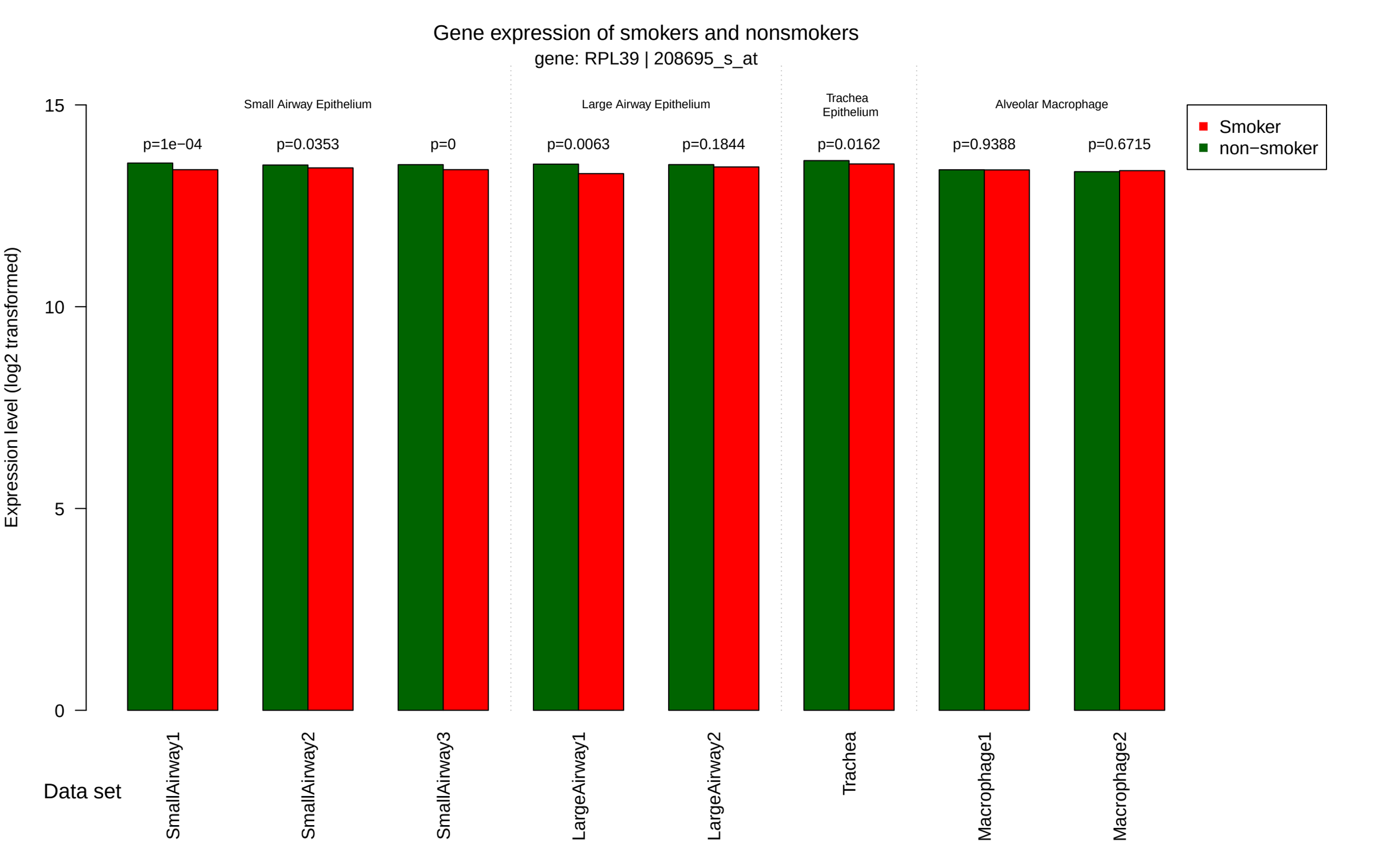
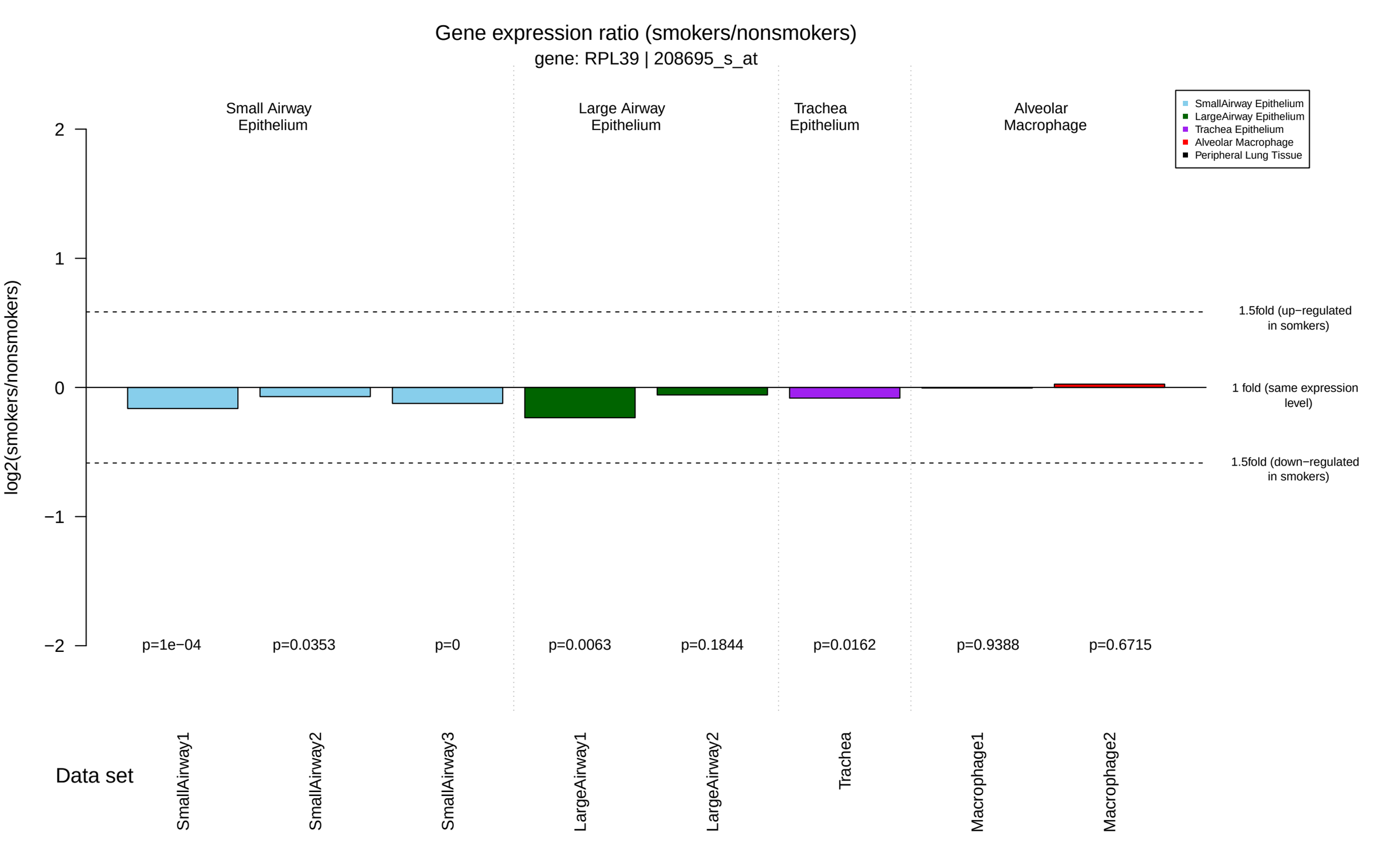
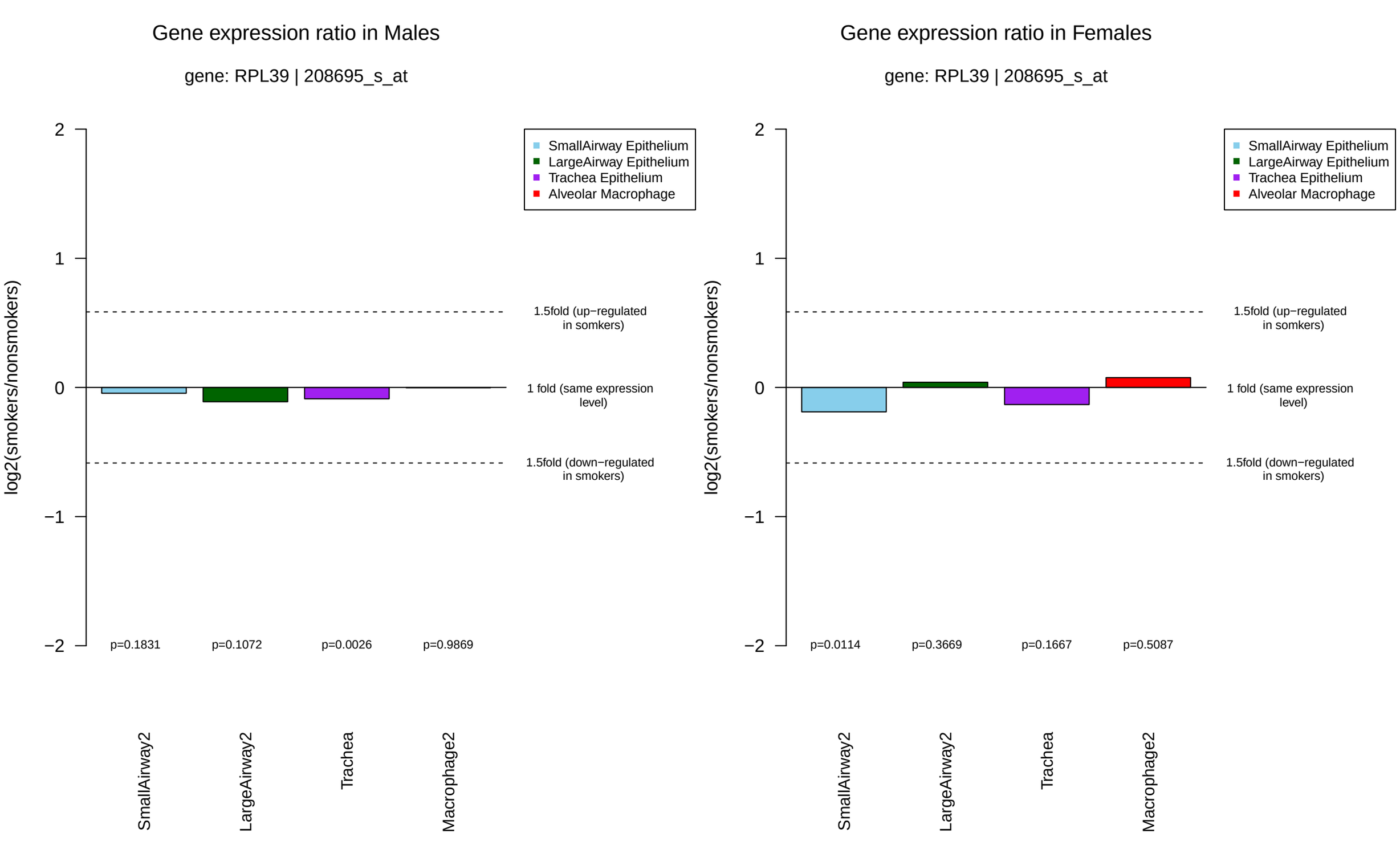
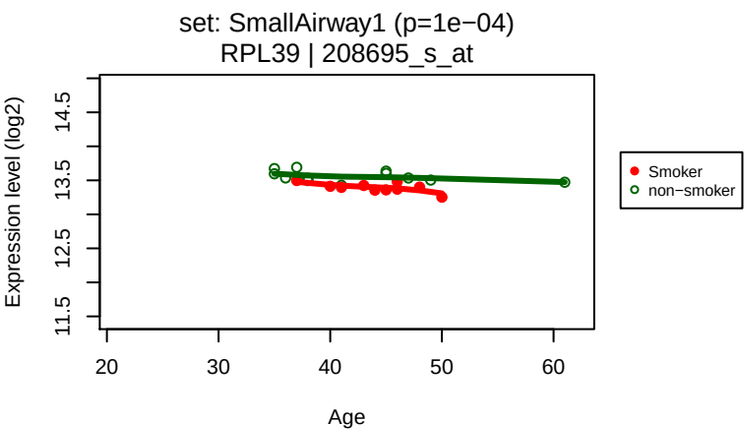
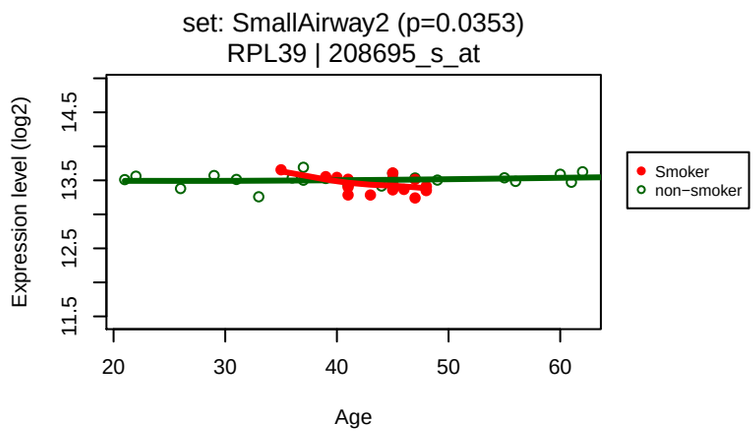
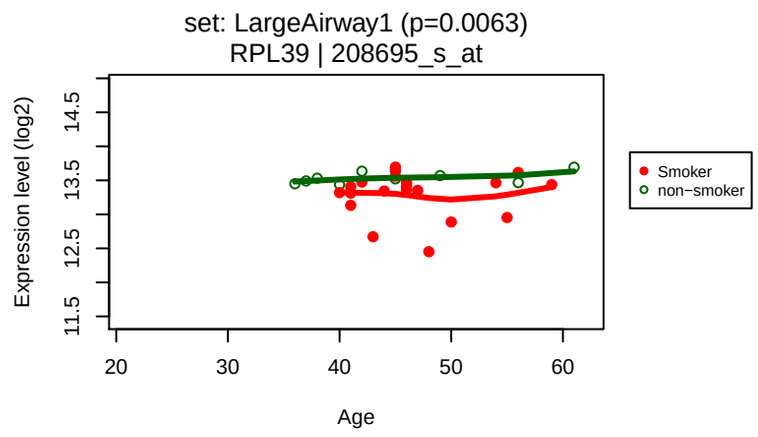

Gene expression of smokers and nonsmokers
gene: RPL39 | 208695_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=1e−04
p=0.0353
p=0
p=0.0063
p=0.1844
p=0.0162
p=0.9388
p=0.6715
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

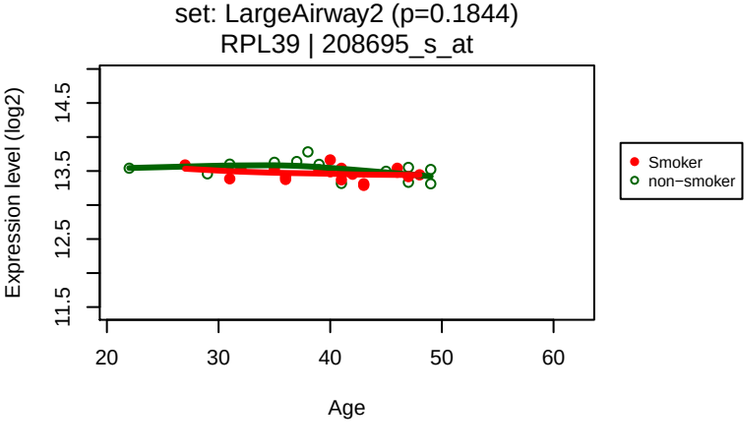

Gene expression ratio (smokers/nonsmokers)
gene: RPL39 | 208695_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0353
p=0
p=0.0063
p=0.1844
p=0.0162
p=0.9388
p=0.6715
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

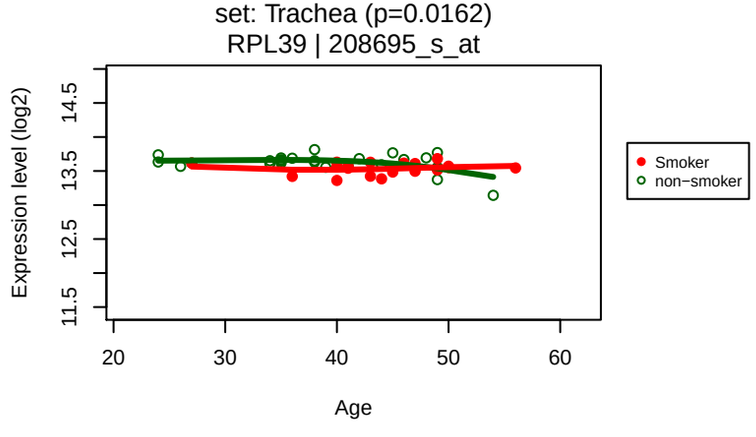

Gene expression ratio in Males
gene: RPL39 | 208695_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1831
p=0.1072
p=0.0026
p=0.9869
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL39 | 208695_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0114
p=0.3669
p=0.1667
p=0.5087
SmallAirway2
LargeAirway2
Trachea
Macrophage2

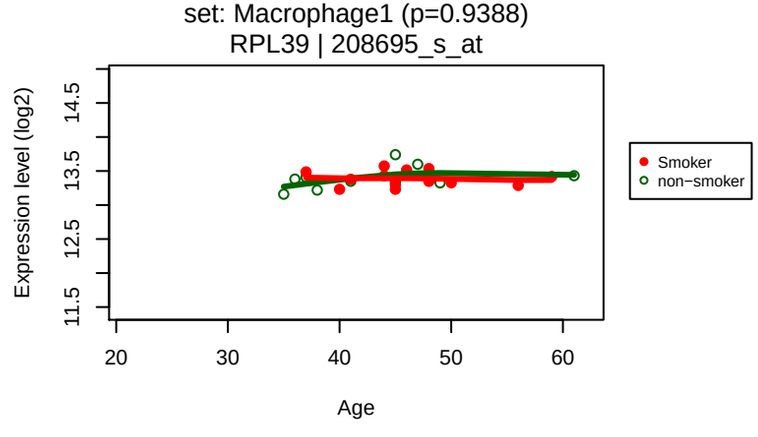

set: SmallAirway1 (p=1e−04)
RPL39 | 208695_s_at
Expression level (log2)
11.5
12.5
13.5
14.5
20
30
40
50
60
Age
Smoker
non−smoker

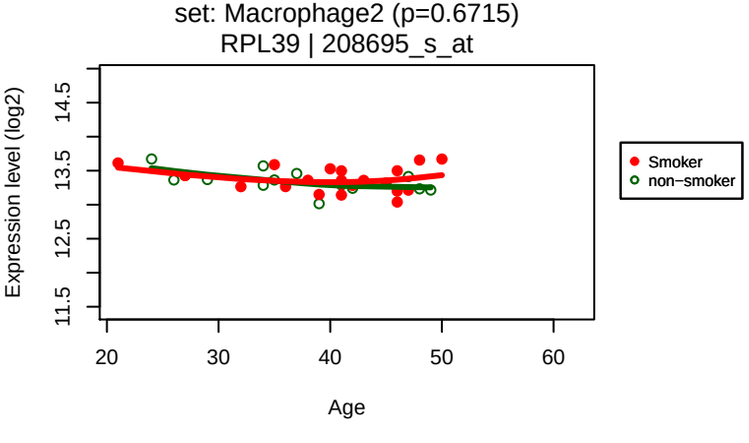

set: SmallAirway2 (p=0.0353)
RPL39 | 208695_s_at
Expression level (log2)
11.5
12.5
13.5
14.5
20
30
40
50
60
Age
Smoker
non−smoker

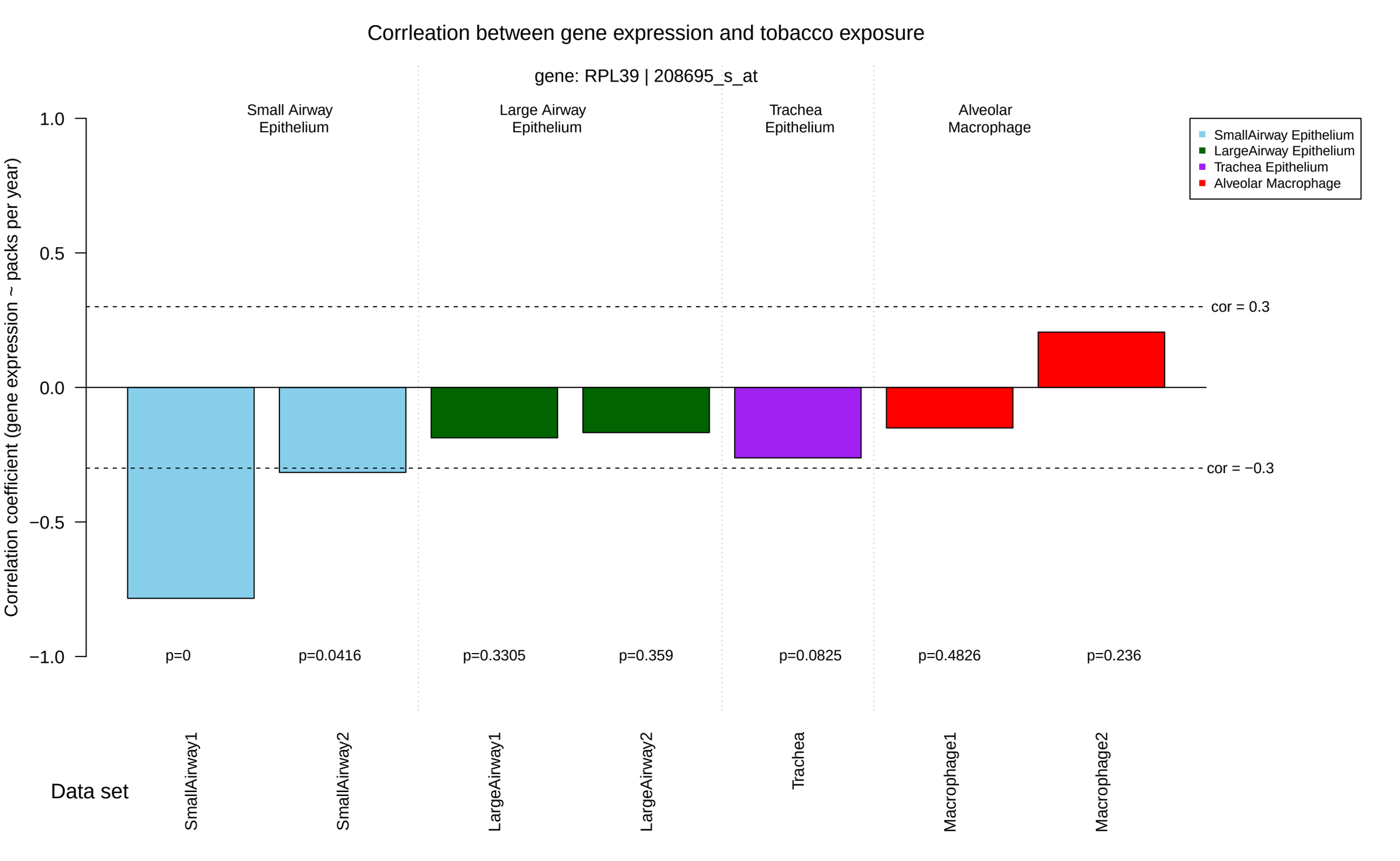

set: LargeAirway1 (p=0.0063)
RPL39 | 208695_s_at
Expression level (log2)
11.5
12.5
13.5
14.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1844)
RPL39 | 208695_s_at
Expression level (log2)
11.5
12.5
13.5
14.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0162)
RPL39 | 208695_s_at
Expression level (log2)
11.5
12.5
13.5
14.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9388)
RPL39 | 208695_s_at
Expression level (log2)
11.5
12.5
13.5
14.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6715)
RPL39 | 208695_s_at
Expression level (log2)
11.5
12.5
13.5
14.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL39 | 208695_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0
p=0.0416
p=0.3305
p=0.359
p=0.0825
p=0.4826
p=0.236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2